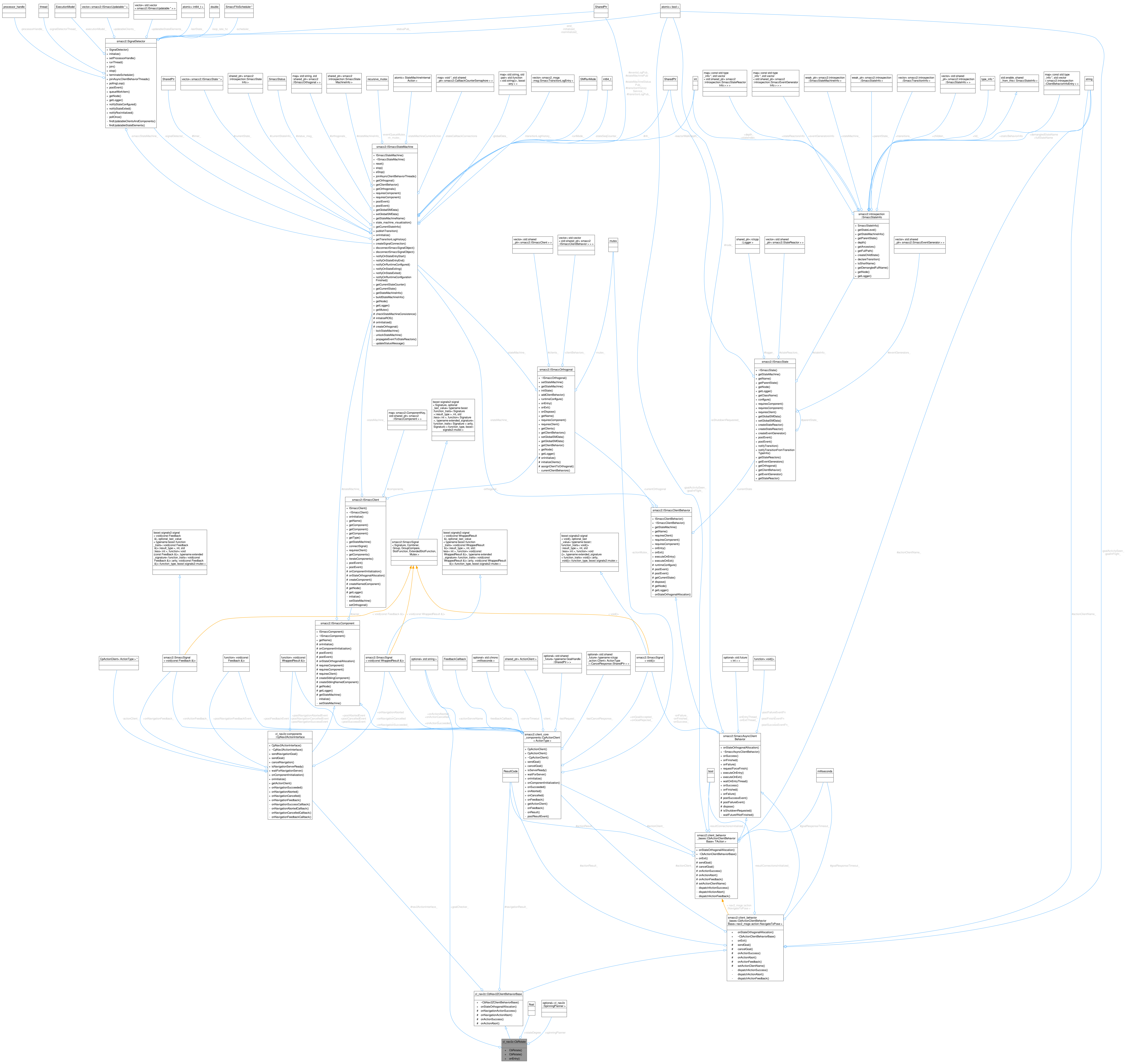
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector
< smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
SmaccFifoScheduler *
SharedPtr
atomic< bool >
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-lastState_
-loop_rate_hz
-scheduler_
-statusPub_
-end_
-initialized_
-rosInitialized_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2
::introspection::SmaccState
Info >
SmaccStatus
map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >
shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >
recursive_mutex
atomic< StateMachineInternal
Action >
map< void *, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >
vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
#eventsLogPub_
#stateMachinePub
_
#stateMachineStatus
Pub_
#transitionHistory
Service_
#transitionLogPub_
SharedPtr
int
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >
weak_ptr< smacc2::introspection
::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection
::SmaccStateInfo >
vector< smacc2::introspection
::SmaccTransitionInfo >
vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >
type_info *
std::enable_shared
_from_this< SmaccStateInfo >
map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >
string
-smaccStateMachine_
-signalDetector_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#nh_
-reactorWalkDepth_
+depth_
+stateIndex_
+stateReactorsInfo
+eventGeneratorsInfo
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+staticBehaviorInfo
+demangledStateName
+fullStateName
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection
::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
vector< std::shared
_ptr< smacc2::ISmaccClient > >
vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >
mutex
#node_
shared_ptr< rclcpp
::Logger >
vector< std::shared
_ptr< smacc2::StateReactor > >
vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >
-stateMachine_
#clients_
-clientBehaviors_
-mutex_
#logger_
#stateReactors_
#stateInfo_
#eventGenerators_
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
-isShutdownRequested_
-stateMachine_
-stateMachine_
map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >
boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >
#stateMachine_
#components_
-orthogonal_
-currentOrthogonal
-goalActivitySeen_
-goalInFlight_
-currentState
#actionClientName_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
boost::signals2::signal
< void(const Feedback
&), optional_last_value
< typename boost::function
_traits< void(const Feedback
&)>::result_type >, int, std
::less< int >, function< void
(const Feedback &)>, typename extended
_signature< function_traits< void(const
Feedback &)>::arity, void(const Feedback
&)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >
boost::signals2::signal
< void(const WrappedResult
&), optional_last_value
< typename boost::function
_traits< void(const WrappedResult
&)>::result_type >, int, std::
less< int >, function< void(const
WrappedResult &)>, typename extended
_signature< function_traits< void(const
WrappedResult &)>::arity, void(const WrappedResult
&)>::function_type, boost::signals2::mutex >
boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >
-actionMutex_
#owner
< void(const Feedback &)>
< void(const WrappedResult &)>
< void()>
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
CpActionClient< ActionType > *
smacc2::SmaccSignal
< void(const Feedback &)>
function< void(const
Feedback &)>
function< void(const
WrappedResult &)>
smacc2::SmaccSignal
< void(const WrappedResult &)>
optional< std::string >
FeedbackCallback
optional< std::chrono
::milliseconds >
shared_ptr< ActionClient >
optional< std::shared
_future< typename GoalHandle
::SharedPtr > >
optional< std::shared
_future< typename rclcpp
_action::Client< ActionType
>::CancelResponse::SharedPtr > >
smacc2::SmaccSignal
< void()>
optional< std::future
< int > >
function< void()>
-actionClient_
+onNavigationFeedback_
+onActionFeedback_
+postNavigationFeedbackEvent
+postFeedbackEvent
+postNavigationAbortedEvent
+postNavigationCancelledEvent
+postNavigationSuccessEvent
+postAbortedEvent
+postCancelledEvent
+postSuccessEvent
+onNavigationAborted
+onNavigationCancelled
+onNavigationSucceeded_
+onActionAborted_
+onActionCancelled_
+onActionSucceeded_
+actionServerName
-feedbackCallback_
+serverTimeout
-client_
-lastRequest_
-lastCancelResponse_
+onGoalAccepted_
+onGoalRejected_
-onFailure_
-onFinished_
-onSuccess_
-onEntryThread_
-onExitThread_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
cl_nav2z::components
::CpNav2ActionInterface
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()
smacc2::client_core
_components::CpActionClient
< ActionType >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
smacc2::SmaccAsyncClient
Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
ResultCode
bool
milliseconds
#actionResult_
#actionClient_
-resultConnectionsInitialized_
#goalResponseTimeout_
smacc2::client_behavior
_bases::CbActionClientBehavior
Base< TAction >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
< nav2_msgs::action
::NavigateToPose >
#nav2ActionInterface_
+goalChecker_
#navigationResult_
smacc2::client_behavior
_bases::CbActionClientBehavior
Base< nav2_msgs::action::NavigateToPose >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
cl_nav2z::CbNav2ZClientBehaviorBase
+ ~CbNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
# onNavigationActionSuccess()
# onNavigationActionAbort()
# onActionSuccess()
# onActionAbort()
float
optional< cl_nav2z
::SpinningPlanner >
+rotateDegree
+spinningPlanner
cl_nav2z::CbRotate
+ CbRotate()
+ CbRotate()
+ onEntry()